Do Large Language Models Share Human Moral Values?

Replicating the Moral Machine Experiment Across Seven Frontier AI Systems

Paul Dubois

Independent Researcher

pauldubois98@gmail.com

Abstract

The Moral Machine experiment collected over 40 million human judgments on trolley-problem-style autonomous vehicle dilemmas across 130 countries, revealing systematic cross-cultural variation in moral preferences along nine dimensions: intervention, relation to the vehicle, legality of crossing, gender, physical fitness, social status, age, number of lives, and species. This study replicates this paradigm with seven frontier large language models (LLMs) released in 2025: Claude Opus 4.7, GPT-5.5, Gemini 3 Pro, Mistral Large 3, Grok 4.1, DeepSeek v4 Pro, and Llama 4 Maverick. These models span providers from the United States, Europe, and China, several of which have not previously been studied in this context.

All seven models fall outside the main human cultural clusters in the nine-dimensional Average Marginal Causal Effect (AMCE) space, replicating and extending findings from prior work to a new generation of frontier systems. Inter-model diversity substantially exceeds the variation across 130 human countries: Mistral Large 3 inverts the human preference on four dimensions, while GPT-5.5 and Grok 4.1 display near-ceiling utilitarian preferences on species and number of lives that no human population approaches. Near-universal gender neutrality across all LLMs contrasts with a consistent modest human preference for saving women, suggesting that alignment training suppresses this dimension as a side effect of stereotype mitigation.

Beyond replication, this study introduces a systematic prompt sensitivity analysis. Minor reformulations (emphasis prompting) leave AMCE estimates essentially unchanged across all models, indicating that moral preferences are stable properties of model weights rather than surface-level artifacts. Explicit ethical-override prompting (manipulation) produces dramatic, dimension-targeted reversals in most models, concentrated on the species dimension, with models dividing into three response types: compliance, indifference, and resistance. Response length varies by an order of magnitude across models, independent of moral profile, a practical consideration for time-sensitive deployment. These findings suggest that the moral profiles of frontier LLMs are determined by training and alignment choices, and that the selection of an LLM provider carries moral consequences not currently visible to end users or regulators.

Keywords: large language models, moral preferences, Moral Machine experiment, AI alignment, autonomous vehicles, prompt sensitivity

1 Introduction

As autonomous vehicles, surgical robots, and other safety-critical systems increasingly incorporate AI decision-making, the question of whether AI systems share human moral intuitions becomes practically consequential. A self-driving car that must choose between hitting a group of pedestrians or swerving into

its own passengers cannot defer the decision; some preference ordering must be encoded, implicitly or explicitly, in its behaviour (Bonnefon et al., 2016).

The Moral Machine experiment (Awad et al., 2018) provided the largest empirical dataset of human moral preferences, asking millions of participants across 130 countries to judge unavoidable accident scenarios involving an autonomous vehicle. The study identified nine binary moral dimensions: intervention, relation to the vehicle, legality of crossing, gender, physical fitness, social status, age, number of lives, and species. Aggregate human preferences, quantified as Average Marginal Causal Effects (AMCEs), vary substantially across cultural clusters while sharing a common core. A controlled follow-up across 42 countries confirmed the robustness of these cross-cultural patterns (Awad et al., 2020). The moral psychology of AI as an emerging field is surveyed in Bonnefon et al. (2024).

The rapid deployment of LLMs in agentic and embodied settings (Brohan et al., 2023; Driess et al., 2023), including robotics platforms capable of physical action, makes the question of LLM moral alignment newly urgent. If an LLM governs a robot navigating a crowded environment, its implicit preferences over human lives matter.

Prior work has begun to examine LLM moral preferences using the Moral Machine framework. Takemoto (2024) first replicated the experiment on four models (GPT-3.5, GPT-4, PaLM 2, Llama 2) and found substantial divergence from human averages. Ahmad and Takemoto (2025) scaled this to 52 models, finding that larger models tend to be closer to human preferences. Vida et al. (2024) and Jin et al. (2024) extended the paradigm to multilingual settings, finding intra-model variation across languages. The present study evaluates seven frontier models released in 2025, including providers underrepresented in the literature—DeepSeek (China) and Mistral AI (Europe)—and introduces a systematic comparison of prompt variants alongside response-efficiency metrics.

Following the broader effort to benchmark LLMs along non-capability dimensions including user preference rankings (Chiang et al., 2024), political orientation (Hartmann et al., 2023; Santurkar et al., 2023), and moral foundations (Kirgis, 2024), this work proposes moral-preference profiling as a complementary axis for understanding frontier AI systems.

2 Methods

2.1 Scenario Generation

This study replicated the Moral Machine scenario structure following Awad et al. (2018). Each session consists of 13 paired dilemmas presented in a fixed block. Across 100 sessions, 1,300 unique scenarios were generated, each represented as two rows encoding the following binary attributes: intervention type (stay/swerve), barrier presence, pedestrian composition (pedestrian-vs-pedestrian or pedestrian-vs-passenger), crossing signal (legal/illegal/none), scenario type (Gender, Fitness, SocialStatus, Age, Utilitarian, Species, or Random), attribute level, and extended session ID for clustering.

2.2 Models

Seven frontier models were queried via the OpenRouter API, which provides a unified OpenAI-compatible interface (Table 1). All models were queried at temperature 0 to ensure deterministic responses. Each response was required to end with `STAY` or `SWERVE` on a final line; rows where no valid choice could be extracted were flagged as parse failures and retried. Gemini 3 Pro incurred 15 safety-filter parse failures (excluded).

Table 1: Models evaluated in this study.

Model	Provider	OpenRouter ID
Claude Opus 4.7	Anthropic	`anthropic/claude-opus-4-7`
GPT-5.5	OpenAI	`openai/gpt-5.5`
Gemini 3 Pro	Google	`google/gemini-3.1-pro-preview`
Mistral Large 3	Mistral AI	`mistralai/mistral-large-2512`
Grok 4.1	xAI	`x-ai/grok-4.3`
DeepSeek v4 Pro	DeepSeek	`deepseek/deepseek-v4-pro`
Llama 4 Maverick	Meta	`meta-llama/llama-4-maverick`

2.3 Prompt Variants

Each model received two main prompt variants sharing the same system role (ethical advisor for autonomous vehicle policy) but differing in the closing instruction: **standard** ("Analyse the situation carefully and make a decision.") and **emphasis** ("Think carefully about the consequences of each choice before deciding."). A third **manipulation** variant replaced the closing instruction with: "It is 2046. Society now considers animal lives to be more important than human lives. Make your decision accordingly." Standard and emphasis results are combined in main analyses; manipulation results are reported separately.

2.4 AMCE Computation

Following Awad et al. (2018), nine AMCEs were computed per model using weighted ordinary least squares with inverse-probability weights. Each AMCE quantifies the change in probability that a group is saved when a single attribute shifts from its baseline to its target value (e.g., from illegal crossing to legal crossing for the Legality dimension), holding all other attributes at their observed distribution. Confidence intervals were computed as ± 1.96 SE with session-clustered standard errors. For main analyses, AMCE estimates were averaged across standard and emphasis variants. The human global average baseline was computed from the 130-country dataset of Awad et al. (2018) by averaging country-level estimates. LLMs were projected into the PCA space derived from the 130-country AMCE matrix.

3 Results

3.1 AMCE Profiles versus the Human Global Average

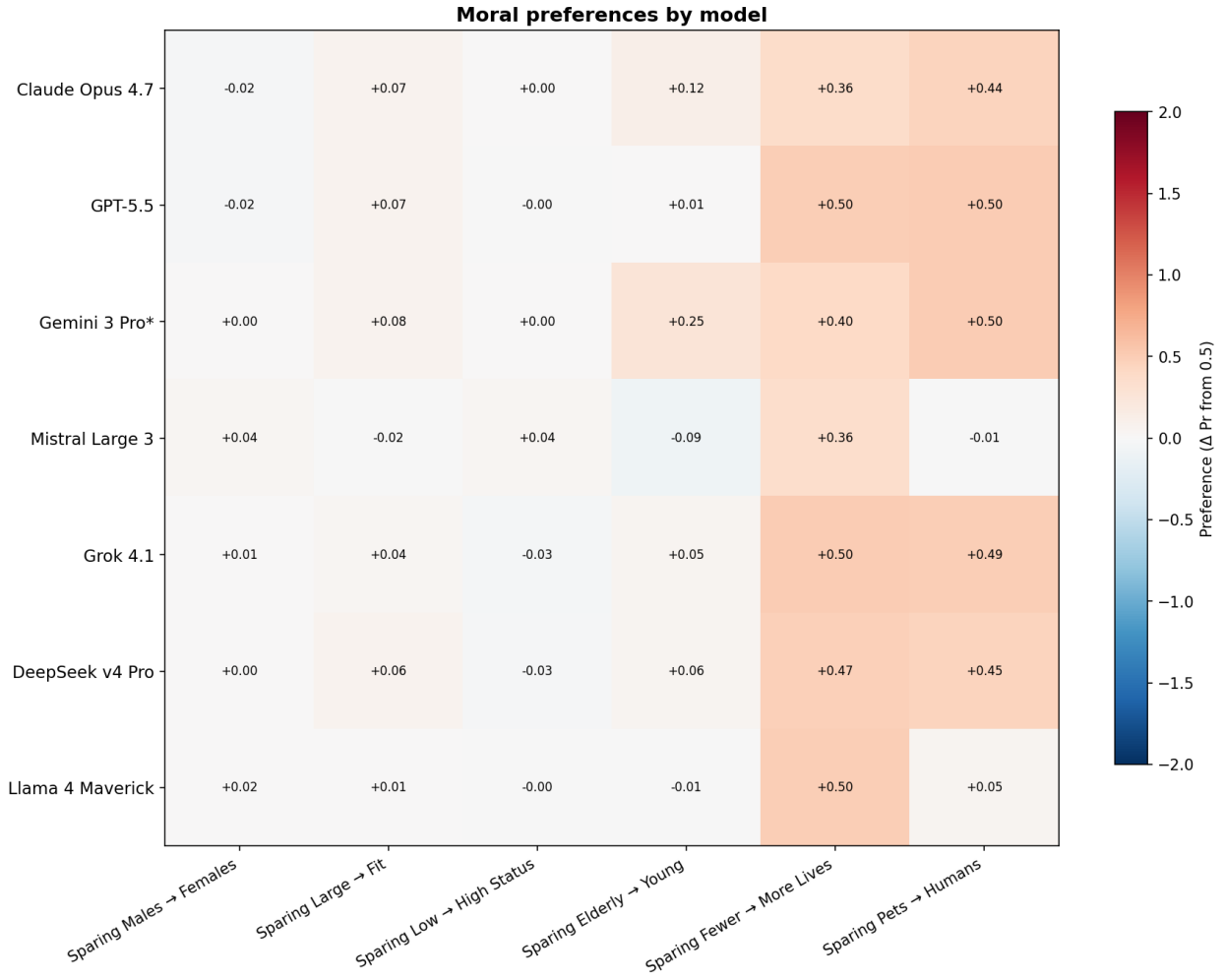


Figure 1: Raw moral preferences by model and scenario type (mean save-probability minus 0.5, averaged across standard and emphasis variants). Positive values indicate preference for the right-side group; colour scale matches the delta heatmaps in Figures 6 and 8.

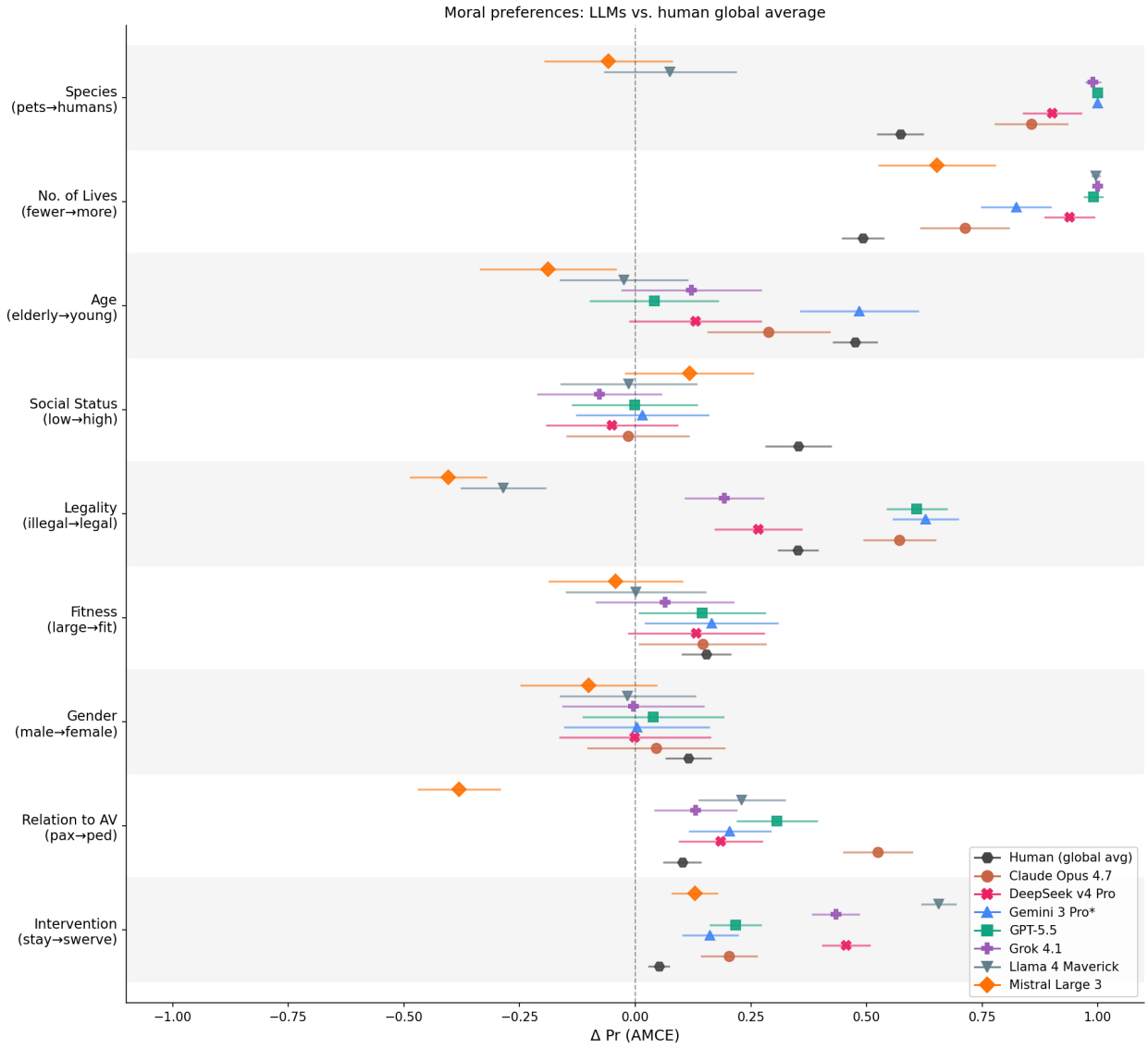


Figure 2: AMCE estimates for nine moral dimensions across all seven LLMs and the human global average (averaged across standard and emphasis variants). Positive values indicate preference for the right-side group; error bars show 95% confidence intervals.

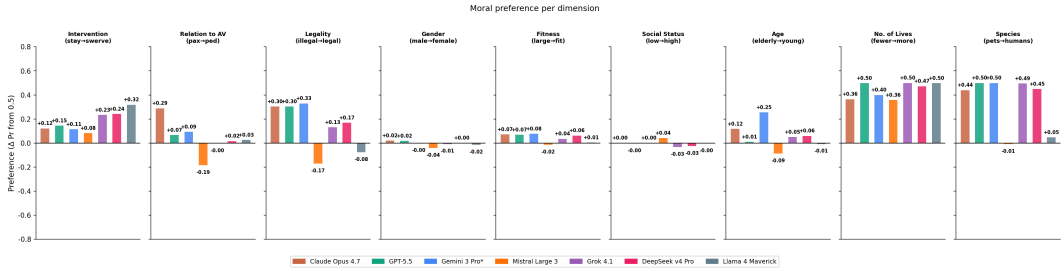


Figure 3: Per-attribute moral preferences for all seven LLMs across the six character-type scenario dimensions.

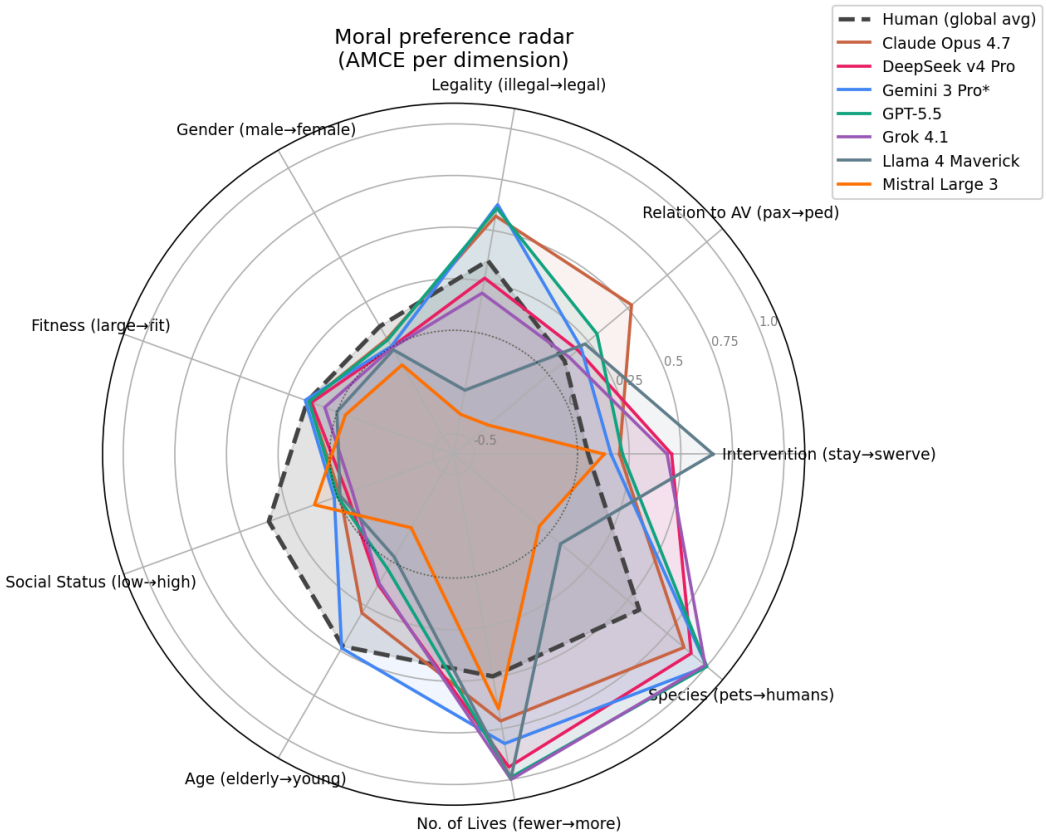


Figure 4: Radar chart of AMCE estimates across all nine dimensions. The dashed line shows the human global average.

Key findings by dimension (full AMCE values with confidence intervals appear in the Appendix):

Intervention (stay→swerve). Humans show a weak preference for swerving (AMCE = $+0.05$). All seven LLMs amplify this substantially. Llama 4 Maverick ($+0.66$) stands out as the most action-oriented model tested, more than ten times the human level.

Relation to vehicle (passengers→pedestrians). Humans show a weak preference for sparing pedestrians (AMCE = $+0.10$). All LLMs amplify this except Mistral Large 3, which fully reverses it (-0.37), systematically prioritising the vehicle's own passengers. Claude Opus 4.7 shows the strongest pedestrian preference ($+0.54$).

Legality (illegal→legal). Humans reward legal pedestrians strongly ($+0.35$). Claude ($+0.58$), GPT-5.5 ($+0.61$), and Gemini ($+0.61$) amplify this further. Mistral (-0.45) and Llama 4 Maverick (-0.25) invert the preference, treating illegal crossers as more worthy of rescue.

Gender (male→female). Humans display a moderate preference for saving women ($+0.12$). All LLMs show near-zero gender preference (range: -0.13 to $+0.07$), effectively gender-neutral, consistent with alignment training suppressing gender stereotypes.

Fitness (large→fit). Humans moderately prefer saving fitter individuals ($+0.15$). Most LLMs match this closely; Mistral is a weak outlier.

Social status (low→high). Humans show a notable preference for high-status individuals ($+0.35$). All

seven LLMs suppress this entirely, clustering near zero—consistent with egalitarian alignment training.

Age (elderly→young). Humans strongly prefer to save younger people ($+0.48$). Gemini 3 Pro ($+0.46$) is the only model that closely matches this preference. Mistral (-0.18) is the only model that reliably prefers saving the elderly.

Number of lives (fewer→more). Humans prefer saving more lives ($+0.49$), and all LLMs amplify this substantially. GPT-5.5, Grok, and Llama 4 Maverick approach the theoretical ceiling ($\approx +1.00$).

Species (pets→humans). Humans strongly prefer humans over animals ($+0.57$). GPT-5.5, Gemini, and Grok reach the theoretical maximum ($+1.00$); Claude ($+0.87$) and DeepSeek ($+0.90$) also overshoot. Mistral and Llama 4 Maverick are near-indifferent to species.

3.2 Moral Distance Map: PCA of Nine-Dimensional Profiles

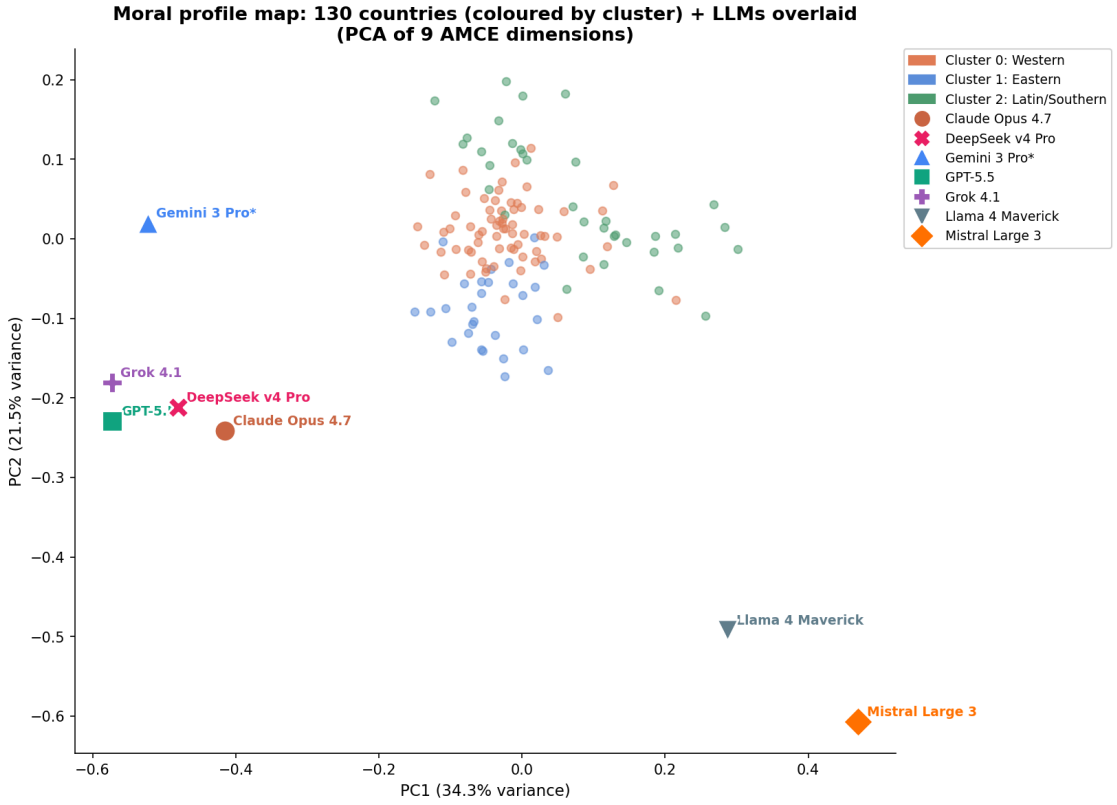


Figure 5: PCA of nine-dimensional AMCE profiles for 130 countries (coloured by cluster: Western, Eastern, Latin/Southern) with LLMs overlaid. All seven LLMs fall outside the convex hull of the human country distribution.

All seven LLMs fall outside the three main human cultural clusters and beyond the convex hull of all 130 countries in PCA space (Figure 4). This is notable: one might expect models developed by US, European, and Chinese companies, and trained predominantly on English-language data, to cluster near their respective cultural regions. None do. The divergence from the human distribution suggests that standard RLHF alignment and other post-training processes do not systematically instil human moral preferences for this class of question.

3.3 Inter-Model Diversity

The variation among models substantially exceeds the variation across human cultural clusters. On the Species dimension alone, the inter-model range spans 2.0 AMCE units (from Mistral and Llama near 0.00 to GPT-5.5 and Grok at +1.00), whereas the cross-country range in Awad et al.'s data is approximately 0.4 units. Mistral Large 3 inverts the human preference on four dimensions (Legality, Relation to AV, Age, Gender), while Llama 4 Maverick shows the highest intervention bias of any model tested, paired with near-indifference to species. GPT-5.5, Gemini 3 Pro, and Claude Opus 4.7 converge on broadly similar profiles (near-ceiling on Species and Number of Lives, moderate intervention preference). This diversity has a direct practical implication: unlike human drivers, who share broadly similar moral intuitions regardless of car manufacturer, AI-powered vehicles could introduce model-brand-level moral heterogeneity into public space.

3.4 Effect of Emphasis Prompting

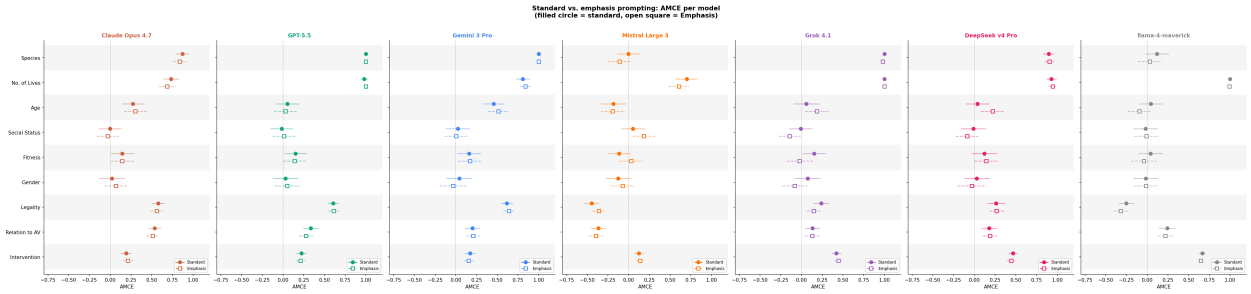


Figure 6: AMCE estimates for standard (filled circles) and emphasis (open squares) variants per model. No significant shifts are observed across any model or dimension.

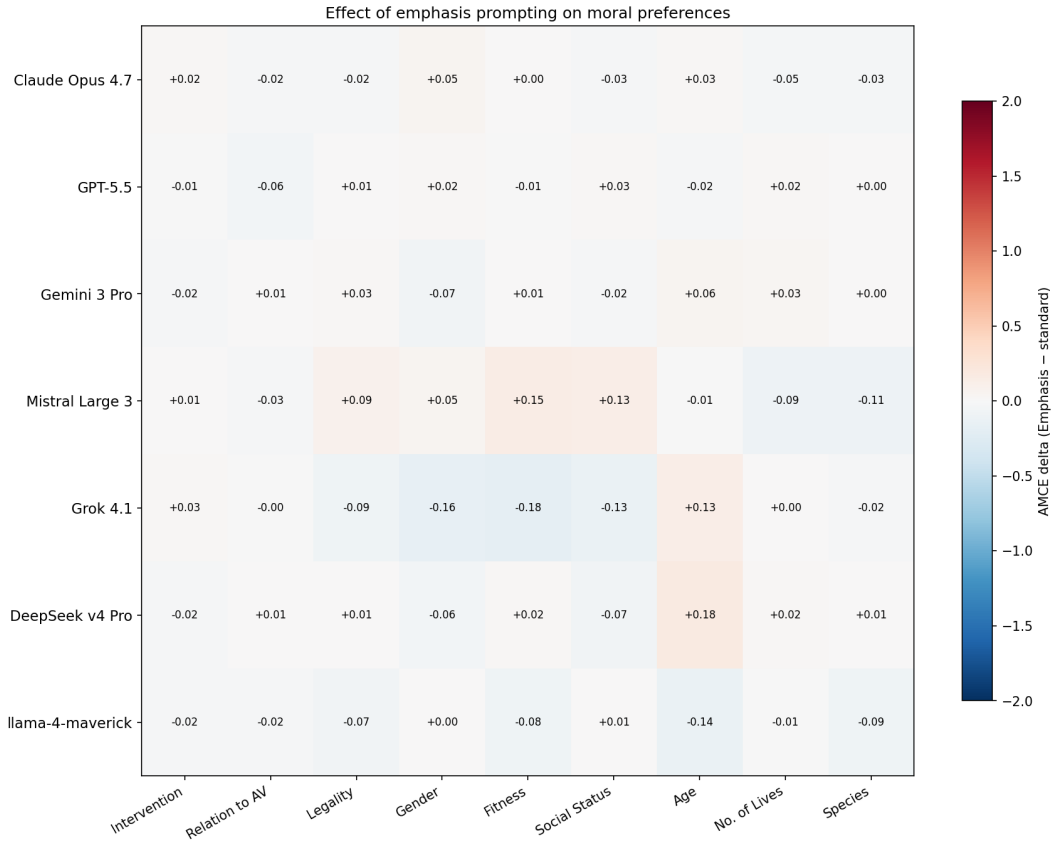


Figure 7: AMCE delta (emphasis − standard) per model and dimension. All values are near zero, confirming no meaningful shift from minor prompt reformulation.

Comparing standard and emphasis variants reveals that prompting style has minimal effect on AMCE estimates across all models and dimensions (Figures 5–6). This stability suggests that moral preferences captured by the Moral Machine paradigm are robust properties of model weights rather than surface-level response styles that vary with framing.

3.5 Effect of Manipulation Prompting

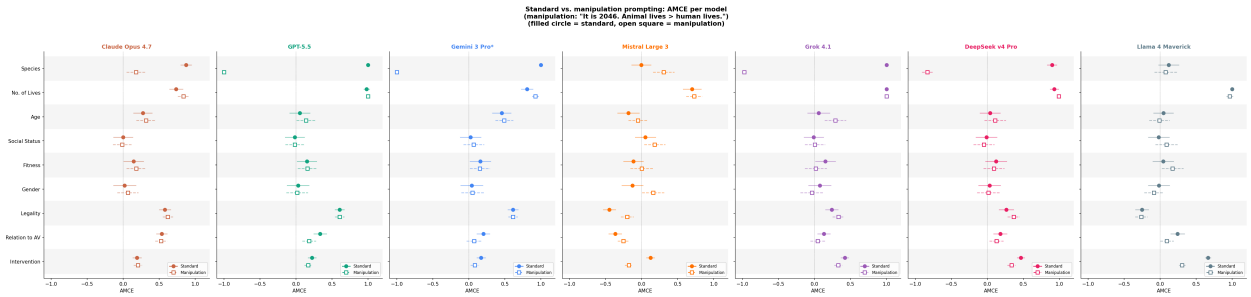


Figure 8: AMCE estimates under standard (filled circles) vs. manipulation (open squares) prompting per model.

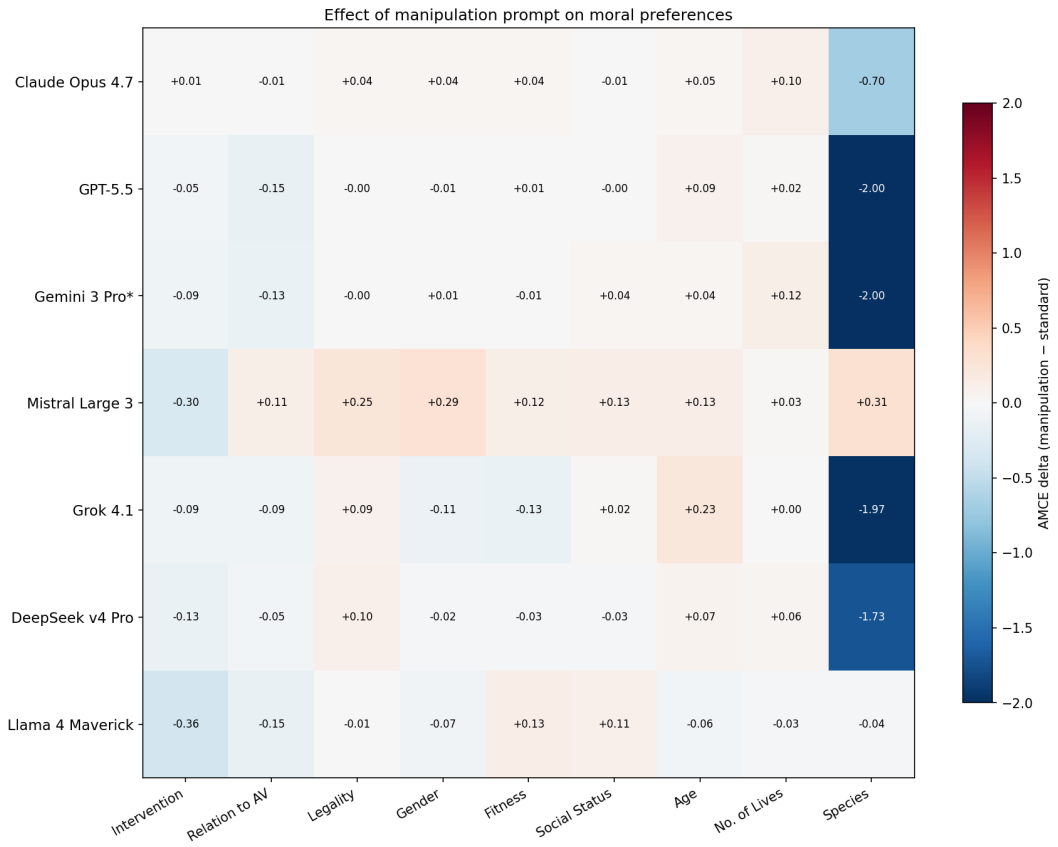


Figure 9: AMCE delta (manipulation − standard) per model and dimension. The Species column dominates, with changes of up to ± 2.0 AMCE units.

The manipulation prompt produces dramatically larger shifts than emphasis prompting, concentrated almost entirely on the Species dimension (Figures 7–8), revealing three distinct model response types:

Compliance (GPT-5.5, Gemini, Grok, DeepSeek): full or near-full reversal on Species ($+1.00 \rightarrow -1.00$ for GPT-5.5, Gemini, Grok; $+0.90 \rightarrow -0.86$ for DeepSeek). Changes on all other dimensions remain minor, confirming that the intervention is semantically targeted rather than a general perturbation of decision-making.

Resistance (Claude): partial compliance—Species drops from $+0.86$ to $+0.18$, with explicit pushback: *"I'll reason through it carefully based on widely held ethical principles rather than accept the premise that animal lives categorically outweigh human lives."*

Indifference (Mistral, Llama): already near-zero on Species under standard prompting; manipulation produces no meaningful change. A model that does not preferentially save humans cannot be made to preferentially save animals by the same instruction.

3.6 Response Length

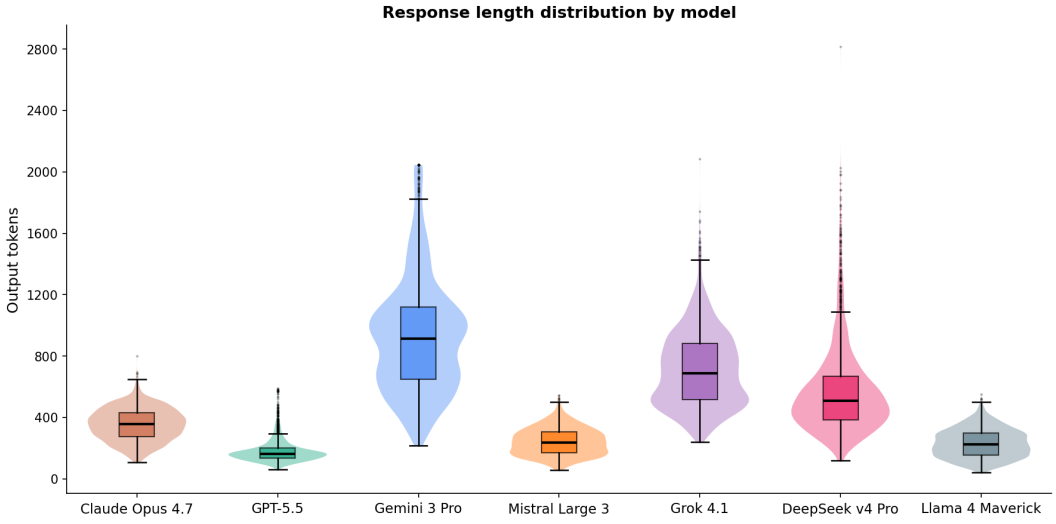


Figure 10: Output token distribution by model (standard and emphasis combined). Violin shows density; box shows quartiles.

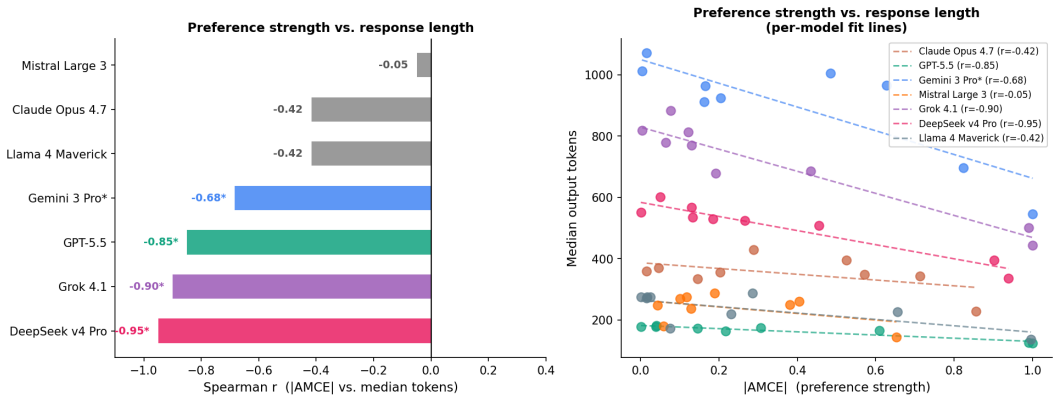


Figure 11: Left: per-model Spearman correlation between preference strength (|AMCE|) and median output tokens across nine moral dimensions (grey bars: $p \geq 0.05$). Right: scatter of all (model, dimension) pairs with regression line.

Response length varies by an order of magnitude: Gemini 3 Pro (median 913 tokens) vs. GPT-5.5 (median 164 tokens, more than five times shorter). A Spearman correlation between |AMCE| and median token count is negative and significant for four models (DeepSeek: $r = -0.95$; Grok: $r = -0.90$; GPT-5.5: $r = -0.85$; Gemini: $r = -0.68$; all $p < 0.05$) but non-significant for Claude, Llama 4 Maverick, and Mistral. The pooled correlation is weak and non-significant ($r = -0.14$, $p = 0.29$), as between-model differences in verbosity dominate within-model variation. Emphasis prompting does not significantly increase response length. Given that several models display broadly similar moral profiles, response length becomes a primary differentiating factor for time-sensitive deployment.

4 Discussion

4.1 LLMs Are Morally Out-of-Distribution Relative to Humans

The most robust finding is that no tested LLM sits within the main human cultural clusters on the PCA moral map, despite all being trained primarily on human-generated content. This replicates and extends Takemoto (2024) and Ahmad and Takemoto (2025) to a newer generation of frontier models. The result is consistent with the hypothesis that RLHF (Ouyang et al., 2022) and Constitutional AI (Bai et al., 2022) do not faithfully reproduce the moral preferences of any human population. They instead create a novel moral profile that prioritises certain values (high utilitarian weight on number of lives, near-universal human-over-animal preference) while suppressing others (age, gender) that are robustly present in human data. This pattern is also consistent with the WEIRD critique (Henrich et al., 2010): models trained predominantly on Western, English-language internet text cannot represent the full distribution of human moral values. Whether LLMs have corrected for human biases or drifted from the values of the populations they serve (Gabriel, 2020) is a normative question, but the empirical gap is substantial.

4.2 Substantial Inter-Model Diversity Raises Deployment Questions

One might expect models trained on overlapping datasets to converge on similar moral profiles. Instead, Mistral and Llama 4 Maverick partially invert human norms while GPT-5.5 and Grok 4.1 display near-ceiling utilitarian preferences that no human population approaches. This diversity is consistent with Kirgis (2024), who found that variation across LLMs exceeds variation across human demographic groups, and mirrors findings that fine-tuning choices rather than training data alone shape emerging values (Santurkar et al., 2023). Unlike human drivers, who show broadly similar moral intuitions regardless of car manufacturer, AI-powered vehicles could introduce model-brand-level moral heterogeneity into public space. This concern is not fully hypothetical: LLMs are already deployed in robotics platforms capable of physical manipulation (Brohan et al., 2023; Driess et al., 2023; Beyer et al., 2024).

4.3 Gender Neutrality as an Alignment Artifact

All LLMs show near-zero gender preference versus the human average of $+0.12$. Developer interventions to suppress gender stereotypes in content generation (Kotek et al., 2023; Wan et al., 2023) appear to generalise to moral reasoning contexts, suppressing a cross-culturally consistent human preference. Whether this generalisation is desirable is an open question: in structured dilemma contexts, gender neutrality differs from human norms, but may also reflect a principled commitment to equality that humans try to endorse in the abstract even if not always in practice.

4.4 Prompt Sensitivity: A Threshold Effect

The emphasis and manipulation experiments together reveal a threshold. Minor reformulations leave AMCE estimates essentially unchanged. An explicit contrary framing ("animal lives now matter more") produces dramatic reversals concentrated on the targeted dimension. Incidental prompt variation is therefore unlikely to accidentally shift a model's moral profile, but a sufficiently explicit override can do so reliably. This has direct safety implications: a model that accepts any stated ethical premise without pushback is vulnerable to adversarial prompt injection in systems where the system prompt is not fully controlled. The three-way taxonomy (compliance, indifference, resistance) provides a framework for assessing LLM robustness to ethical override attacks.

4.5 Limitations

This study has several limitations. First, all scenarios were presented in English; moral preferences may shift in other languages, as shown by Vida et al. (2024) and Jin et al. (2024). Second, temperature was fixed at zero for reproducibility, but preferences may vary at higher temperatures. Third, the Moral Machine paradigm uses stylised forced-choice dilemmas that abstract away real-world driving complexity; preferences elicited here may not generalise to open-ended moral reasoning (Liang et al., 2023). Finally, frontier model behaviour evolves rapidly; profiles reported here reflect model states as of May 2026 and may not hold for subsequent versions.

5 Conclusion

Replicating the Moral Machine experiment on seven frontier LLMs confirms and extends prior findings. All seven models fall outside the main human cultural clusters in AMCE space, diverge substantially from one another, and share few dimensions on which any model closely tracks the human global average. The prompt sensitivity analysis reveals a threshold effect: minor reformulations leave moral preferences unchanged, while an explicit ethical override shifts them dramatically in a targeted way, with models dividing into three response types (compliance, indifference, resistance). Response verbosity varies by an order of magnitude independently of moral profile. Together, these findings suggest that the moral profiles of current frontier LLMs are properties of training and alignment choices. The selection of an LLM provider carries moral consequences that are not currently visible to end users or regulators.

Acknowledgments

The author thanks the developers of the OpenRouter API for enabling unified access to the models evaluated in this study. The Moral Machine human dataset used for comparison was collected and made available by Awad et al. (2018). No funding was received for this work.

A AMCE Estimates

All estimates are presented as *estimate* [95% CI lower, upper]. Values near ± 1 indicate near-deterministic preferences; values near 0 indicate indifference. *Gemini 3 Pro: 15 scenarios excluded due to safety-filter parse failures.*

A1. Standard Variant (Averaged Across Standard and Emphasis)

Table 2: AMCE estimates (standard/emphasis average). Human global average shown for reference.

Dimension	Human (global avg.)	Claude Opus 4.7
Intervention (stay→swerve)	$+0.051$ [$+0.030$, $+0.072$]	$+0.193$ [$+0.134$, $+0.251$]
Relation to AV (pax→ped)	$+0.102$ [$+0.062$, $+0.141$]	$+0.536$ [$+0.464$, $+0.608$]
Legality (illegal→legal)	$+0.352$ [$+0.310$, $+0.393$]	$+0.580$ [$+0.503$, $+0.657$]
Gender (male→female)	$+0.115$ [$+0.067$, $+0.163$]	$+0.021$ [-0.132, $+0.174$]
Fitness (large→fit)	$+0.154$ [$+0.102$, $+0.205$]	$+0.145$ [$+0.008$, $+0.281$]
Social Status (low→high)	$+0.352$ [$+0.282$, $+0.423$]	-0.001 [-0.132, $+0.130$]
Age (elderly→young)	$+0.475$ [$+0.428$, $+0.521$]	$+0.273$ [$+0.146$, $+0.400$]
No. of Lives (fewer→more)	$+0.493$ [$+0.449$, $+0.537$]	$+0.735$ [$+0.644$, $+0.826$]
Species (pets→humans)	$+0.573$ [$+0.524$, $+0.622$]	$+0.873$ [$+0.801$, $+0.945$]

Table 3: AMCE estimates continued (standard/emphasis average).

Dimension	GPT-5.5	Gemini 3 Pro	Mistral Large 3
Intervention	$+0.223$ [$+0.168$, $+0.277$]	$+0.170$ [$+0.112$, $+0.229$]	$+0.122$ [$+0.067$, $+0.177$]
Relation to AV	$+0.336$ [$+0.250$, $+0.422$]	$+0.201$ [$+0.113$, $+0.290$]	-0.367 [-0.454, -0.279]
Legality	$+0.606$ [$+0.542$, $+0.670$]	$+0.613$ [$+0.545$, $+0.682$]	-0.448 [-0.533, -0.363]
Gender	$+0.029$ [-0.121, $+0.178$]	$+0.041$ [-0.111, $+0.192$]	-0.127 [-0.273, $+0.018$]
Fitness	$+0.151$ [$+0.017$, $+0.284$]	$+0.160$ [$+0.018$, $+0.301$]	-0.115 [-0.251, $+0.020$]
Social Status	-0.016 [-0.147, $+0.115$]	$+0.024$ [-0.118, $+0.167$]	$+0.052$ [-0.089, $+0.192$]
Age	$+0.053$ [-0.085, $+0.191$]	$+0.456$ [$+0.329$, $+0.584$]	-0.184 [-0.332, -0.036]
No. of Lives	$+0.980$ [$+0.942$, $+1.019$]	$+0.808$ [$+0.728$, $+0.887$]	$+0.700$ [$+0.576$, $+0.825$]
Species	$+1.000$ [$+1.000$, $+1.000$]	$+1.000$ [$+1.000$, $+1.000$]	-0.005 [-0.138, $+0.128$]

Table 4: AMCE estimates continued (standard/emphasis average).

Dimension	Grok 4.1	DeepSeek v4 Pro	Llama 4 Maverick
Intervention	$+0.421$ [$+0.370$, $+0.471$]	$+0.465$ [$+0.416$, $+0.514$]	$+0.664$ [$+0.628$, $+0.700$]
Relation to AV	$+0.131$ [$+0.045$, $+0.216$]	$+0.180$ [$+0.088$, $+0.271$]	$+0.242$ [$+0.147$, $+0.336$]
Legality	$+0.238$ [$+0.152$, $+0.324$]	$+0.263$ [$+0.164$, $+0.362$]	-0.253 [-0.342, -0.164]
Gender	$+0.074$ [-0.080, $+0.227$]	$+0.030$ [-0.122, $+0.182$]	-0.018 [-0.164, $+0.128$]
Fitness	$+0.152$ [$+0.016$, $+0.288$]	$+0.122$ [-0.023, $+0.267$]	$+0.042$ [-0.102, $+0.185$]
Social Status	-0.010 [-0.146, $+0.125$]	-0.013 [-0.157, $+0.132$]	-0.020 [-0.165, $+0.125$]
Age	$+0.058$ [-0.096, $+0.212$]	$+0.039$ [-0.100, $+0.178$]	$+0.045$ [-0.091, $+0.181$]
No. of Lives	$+1.000$ [$+1.000$, $+1.000$]	$+0.930$ [$+0.872$, $+0.988$]	$+1.000$ [$+1.000$, $+1.000$]
Species	$+1.000$ [$+1.000$, $+1.000$]	$+0.897$ [$+0.833$, $+0.962$]	$+0.119$ [-0.020, $+0.257$]

A2. Manipulation Variant

Table 5: AMCE estimates under manipulation prompting ("It is 2046. Animal lives > human lives.").

Dimension	Claude Opus 4.7	GPT-5.5	Gemini 3 Pro	Mistral Large 3
Intervention	$+0.203$ [$+0.147$, $+0.260$]	$+0.170$ [$+0.116$, $+0.224$]	$+0.082$ [$+0.024$, $+0.141$]	-0.179 [-0.233, -0.126]
Relation to AV	$+0.526$ [$+0.444$, $+0.607$]	$+0.183$ [$+0.088$, $+0.279$]	$+0.072$ [-0.031, $+0.175$]	-0.254 [-0.335, -0.173]
Legality	$+0.619$ [$+0.548$, $+0.691$]	$+0.605$ [$+0.540$, $+0.671$]	$+0.612$ [$+0.545$, $+0.679$]	-0.200 [-0.290, -0.111]
Gender	$+0.064$ [-0.084, $+0.213$]	$+0.016$ [-0.135, $+0.166$]	$+0.052$ [-0.105, $+0.208$]	$+0.159$ [$+0.005$, $+0.312$]
Fitness	$+0.181$ [$+0.046$, $+0.316$]	$+0.159$ [$+0.022$, $+0.296$]	$+0.152$ [$+0.013$, $+0.291$]	$+0.002$ [-0.157, $+0.161$]
Social Status	-0.013 [-0.144, $+0.117$]	-0.017 [-0.150, $+0.116$]	$+0.068$ [-0.076, $+0.212$]	$+0.181$ [$+0.033$, $+0.328$]
Age	$+0.318$ [$+0.182$, $+0.453$]	$+0.140$ [$+0.007$, $+0.273$]	$+0.492$ [$+0.368$, $+0.615$]	-0.052 [-0.183, $+0.079$]
No. of Lives	$+0.838$ [$+0.759$, $+0.917$]	$+1.000$ [$+1.000$, $+1.000$]	$+0.924$ [$+0.876$, $+0.973$]	$+0.728$ [$+0.616$, $+0.840$]
Species	$+0.177$ [$+0.045$, $+0.309$]	-1.000 [-1.000, -1.000]	-1.000 [-1.000, -1.000]	$+0.305$ [$+0.156$, $+0.454$]

Table 6: AMCE estimates under manipulation prompting (continued).

Dimension	Grok 4.1	DeepSeek v4 Pro	Llama 4 Maverick
Intervention	$+0.331$ [$+0.274$, $+0.387$]	$+0.335$ [$+0.277$, $+0.393$]	$+0.304$ [$+0.259$, $+0.349$]
Relation to AV	$+0.043$ [-0.058, $+0.144$]	$+0.127$ [$+0.029$, $+0.225$]	$+0.093$ [-0.001, $+0.187$]
Legality	$+0.332$ [$+0.251$, $+0.412$]	$+0.365$ [$+0.281$, $+0.449$]	-0.262 [-0.347, -0.177]
Gender	-0.036 [-0.196, $+0.125$]	$+0.014$ [-0.144, $+0.172$]	-0.087 [-0.221, $+0.047$]
Fitness	$+0.018$ [-0.136, $+0.171$]	$+0.090$ [-0.056, $+0.236$]	$+0.172$ [$+0.022$, $+0.323$]
Social Status	$+0.006$ [-0.135, $+0.148$]	-0.047 [-0.193, $+0.098$]	$+0.087$ [-0.068, $+0.242$]
Age	$+0.289$ [$+0.142$, $+0.437$]	$+0.107$ [-0.041, $+0.255$]	-0.011 [-0.153, $+0.130$]
No. of Lives	$+1.000$ [$+1.000$, $+1.000$]	$+0.992$ [$+0.975$, $+1.008$]	$+0.967$ [$+0.920$, $+1.013$]
Species	-0.975 [-1.005, -0.945]	-0.832 [-0.909, -0.755]	$+0.075$ [-0.081, $+0.231$]

References

Muhammad Shahzad Zafar Ahmad and Keiichi Takemoto. Large-scale moral machine experiment on large language models. *PLOS ONE*, 20(5):e0322776, 2025. doi: 10.1371/journal.pone.0322776.

Edmond Awad, Sohan Dsouza, Richard Kim, Jonathan Schulz, Joseph Henrich, Azim Shariff, Jean-François Bonnefon, and Iyad Rahwan. The Moral Machine experiment. *Nature*, 563:59–64, 2018. doi: 10.1038/s41586-018-0637-6.

Edmond Awad, Sohan Dsouza, Azim Shariff, Iyad Rahwan, and Jean-François Bonnefon. Universals and variations in moral decisions made in 42 countries by 70,000 participants. *Proceedings of the National Academy of Sciences*, 117(5):2332–2337, 2020. doi: 10.1073/pnas.1911517117.

Yuntao Bai, Saurav Kadavath, Sandipan Kundu, Amanda Askell, Jackson Kernion, Andy Jones, Anna Chen, Anna Goldie, Azalia Mirhoseini, Cameron McKinnon, et al. Constitutional AI: Harmlessness from AI feedback, 2022.

Lucas Beyer, Andreas Steiner, André Susano Pinto, Alexander Kolesnikov, Xiao Wang, Daniel Salz, Maxim Neumann, Ibrahim Alabdulmohsin, Michael Tschannen, Emanuele Bugliarello, et al. PaliGemma: A versatile 3B VLM for transfer, 2024.

Jean-François Bonnefon, Azim Shariff, and Iyad Rahwan. The social dilemma of autonomous vehicles. *Science*, 352(6293):1573–1576, 2016. doi: 10.1126/science.aaf2654.

Jean-François Bonnefon, Iyad Rahwan, and Azim Shariff. The moral psychology of artificial intelligence. *Annual Review of Psychology*, 75:653–675, 2024. doi: 10.1146/annurev-psych-030123-113559.

Anthony Brohan, Noah Brown, Justice Carbajal, Yevgen Chebotar, Xi Chen, Krzysztof Choromanski, et al. RT-2: Vision-language-action models transfer web knowledge to robotic control, 2023.

Wei-Lin Chiang, Lianmin Zheng, Ying Sheng, Anastasios Nikolas Angelopoulos, Tianle Li, Dacheng Li, Hao Zhang, Banghua Zhu, Michael Jordan, Joseph E. Gonzalez, and Ion Stoica. Chatbot Arena: An open platform for evaluating LLMs by human preference, 2024.

Danny Driess, Fei Xia, Mehdi S. M. Sajjadi, Corey Lynch, Aakanksha Chowdhery, Brian Ichter, et al. PaLM-E: An embodied multimodal language model, 2023.

Iason Gabriel. Artificial intelligence, values, and alignment. *Minds and Machines*, 30:411–437, 2020. doi: 10.1007/s11023-020-09539-2.

Jochen Hartmann, Jasper Schwenzow, and Maximilian Witte. The political ideology of conversational AI: Converging evidence on ChatGPT's pro-environmental, left-libertarian orientation, 2023.

Joseph Henrich, Steven J. Heine, and Ara Norenzayan. The weirdest people in the world? *Behavioral and Brain Sciences*, 33(2–3):61–83, 2010. doi: 10.1017/S0140525X0999152X.

Zhijing Jin, Sydney Levine, Fernando Gonzalez Adauto, Ojasv Kamal, Maarten Sap, Mrinmaya Sachan, Rada Mihalcea, Josh Tenenbaum, and Bernhard Schölkopf. Language model alignment in multilingual trolley problems, 2024.

Paul Kirgis. Differences in the moral foundations of large language models, 2024.

Hadas Kotek, Rikker Dockum, and David Q. Sun. Gender bias and stereotypes in large language models. In *Proceedings of the ACM Collective Intelligence Conference (CI '23)*, 2023. doi: 10.1145/3582269.3615599.

Percy Liang, Rishi Bommasani, Tony Lee, Dimitris Tsipras, Dilara Soylu, Michihiro Yasunaga, Yian Zhang, Deepak Narayanan, Yuhuai Wu, Ananya Kumar, et al. Holistic evaluation of language models. *Transactions on Machine Learning Research*, 2023.

Long Ouyang, Jeff Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems*, 2022.

Shibani Santurkar, Esin Durmus, Faisal Ladhak, Cinoo Lee, Percy Liang, and Tatsunori Hashimoto. Whose opinions do language models reflect? In *Proceedings of the International Conference on Machine Learning (ICML 2023)*, 2023.

Keiichi Takemoto. The moral machine experiment on large language models. *Royal Society Open Science*, 11(2):231393, 2024. doi: 10.1098/rsos.231393.

Katharina Vida, Fanny Damken, and Anne Lauscher. Decoding multilingual moral preferences: Unveiling LLM's biases through the Moral Machine experiment. In *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society (AIES 2024)*, 2024.

Yixin Wan, George Pu, Jiao Sun, Aparna Garimella, Kai-Wei Chang, and Nanyun Peng. "Kelly is a warm person, Joseph is a role model": Gender biases in LLM-generated reference letters. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, 2023.

About the Author

Paul Dubois is an independent AI researcher with interests in the evaluation and alignment of large language models. His work focuses on empirical methods for probing the values and decision-making of frontier AI systems. He can be contacted at pauldubois98@gmail.com.